H1000 Hardened S45500 Stainless Steel

H1000 hardened S45500 stainless steel is S45500 stainless steel in the hardened (H) condition. The graph bars on the material properties cards below compare H1000 hardened S45500 stainless steel to: wrought precipitation-hardening stainless steels (top), all iron alloys (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Brinell Hardness

410

Elastic (Young's, Tensile) Modulus

190 GPa 28 x 10^6 psi

Elongation at Break

11 %

Fatigue Strength

780 MPa 110 x 10^3 psi

Poisson's Ratio

0.28

Reduction in Area

45 %

Rockwell C Hardness

46

Shear Modulus

75 GPa 11 x 10^6 psi

Shear Strength

970 MPa 140 x 10^3 psi

Tensile Strength: Ultimate (UTS)

1610 MPa 230 x 10^3 psi

Tensile Strength: Yield (Proof)

1430 MPa 210 x 10^3 psi

Thermal Properties

Latent Heat of Fusion

270 J/g

Maximum Temperature: Corrosion

620 °C 1140 °F

Maximum Temperature: Mechanical

760 °C 1400 °F

Melting Completion (Liquidus)

1440 °C 2630 °F

Melting Onset (Solidus)

1400 °C 2550 °F

Specific Heat Capacity

470 J/kg-K 0.11 BTU/lb-°F

Thermal Expansion

11 µm/m-K

Otherwise Unclassified Properties

Base Metal Price

17 % relative

Density

7.9 g/cm^3 490 lb/ft^3

Embodied Carbon

3.8 kg CO_2/kg material

Embodied Energy

57 MJ/kg 24 x 10^3 BTU/lb

Embodied Water

120 L/kg 14 gal/lb

Common Calculations

PREN (Pitting Resistance)

13

Resilience: Ultimate (Unit Rupture Work)

180 MJ/m^3

Resilience: Unit (Modulus of Resilience)

5320 kJ/m^3

Stiffness to Weight: Axial

14 points

Stiffness to Weight: Bending

24 points

Strength to Weight: Axial

57 points

Strength to Weight: Bending

39 points

Thermal Shock Resistance

56 points

Alloy Composition

Iron (Fe)	71.5 to 79.2
Chromium (Cr)	11 to 12.5
Nickel (Ni)	7.5 to 9.5
Copper (Cu)	1.5 to 2.5
Titanium (Ti)	0.8 to 1.4
Manganese (Mn)	0 to 0.5
Silicon (Si)	0 to 0.5
Molybdenum (Mo)	0 to 0.5
Niobium (Nb)	0 to 0.5
Tantalum (Ta)	0 to 0.5
Carbon (C)	0 to 0.050
Phosphorus (P)	0 to 0.040
Sulfur (S)	0 to 0.030

All values are % weight. Ranges represent what is permitted under applicable standards.

Followup Questions

How are the material properties defined?

How do I compare this exact variant to another material?

Further Reading

ASTM A564: Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

Welding Metallurgy of Stainless Steels, Erich Folkhard et al., 2012

ASTM A959: Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels

Properties and Selection: Irons, Steels and High Performance Alloys, ASM Handbook vol. 1, ASM International, 1993

Corrosion of Stainless Steels, A. John Sedriks, 1996

ASM Specialty Handbook: Stainless Steels, J. R. Davis (editor), 1994

Advances in Stainless Steels, Baldev Raj et al. (editors), 2010